

Tribhuvan University
Institute of Science and Technology
SCHOOL OF MATHEMATICAL SCIENCES
Second Assessment 2083

Subject: **Research Methodology**
Course No: **MDS 601**
Level: **MDS / II Year / III Semester**

Full Marks: 45
Pass Marks: 22.5
Time: 2 hrs

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A [5 x 3 = 15]

1. Define literature review. What are the benefits of a good literature survey?
2. Explain ordinal and ratio scale with suitable examples.
3. What is case study research? Describe the characteristics of case study research.
4. What is bias? How can it be reduced while interviewing respondents?
5. Explain how stratified sampling can reduce the standard error.

Group B [5 x 6 = 30]

6. A manager notices that the number of grievances in the organization is increasing. The manager wishes to investigate this occurrence. What research design seems appropriate for the study? Give appropriate explanation.
7. Define the terms reliability and validity. Describe the various types of validity.
8. Describe the APA method of citation for single author and multiple authors with examples.
9. Define sampling. Why is sampling used in research? List the principal steps involved in the process of sampling.

OR

Distinguish between sampling error and non-sampling error. How can non-sampling error be reduced in the process of data collection?

10. A researcher wants to conduct a survey where the population proportion is estimated to be 10%, the desired error of estimation is 2%, and the level of significance is 5%. Determine the required sample size.

OR

What is a research proposal? Explain the contents and structure of a research proposal with brief explanation.
